Viglet Turing ES

Administration Guide

Viglet Team

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Preface

Viglet Turing ES (https://viglet.com/turing) is an open source solution (https://github.com/openturing), which has Semantic Navigation and Chatbot as its main features. You can choose from several NLPs to enrich the data. All content is indexed in Solr as search engine.

Chapter 1. Architecture

[Architecture] | *img/turing-diagram.png*

Figure 1. Turing ES Architecture

Chapter 2. NLP

Turing support the followings providers:

2.1. OpenNLP

Apache OpenNLP is a machine learning based toolkit for the processing of natural language text.

Website: https://opennlp.apache.org/

2.2. OpenText Content Analytics

It transforms data into insights for better decision-making and information management while freeing up resources and time.

Website: https://www.opentext.com/

2.3. CoreNLP

CoreNLP is your one stop shop for natural language processing in Java! CoreNLP enables users to derive linguistic annotations for text, including token and sentence boundaries, parts of speech, named entities, numeric and time values, dependency and constituency parses, coreference, sentiment, quote attributions, and relations. CoreNLP currently supports 6 languages: Arabic, Chinese, English, French, German, and Spanish.

Website: https://stanfordnlp.github.io/CoreNLP/,

2.4. SpaCy

It is a free open-source library for Natural Language Processing in Python. It features NER, POS tagging, dependency parsing, word vectors and more.

Website: https://spacy.io

2.5. Polyglot NLP

Polyglot is a natural language pipeline that supports massive multilingual applications.

Website: https://polyglot.readthedocs.io

Chapter 3. Documents and OCR

It can read PDFs and Documents and convert to plain text and also it uses OCR to detect text in images and images into documents.

Chapter 4. Semantic Navigation

4.1. Connectors

Semantic Navigation uses Connectors to index the content from many sources.

4.1.1. Apache Nutch

Plugin for Apache Nutch to index content using crawler.

Learn more at https://docs.viglet.com/turing/connectors/#nutch

4.1.2. Database

Command line that uses the same concept as sqoop (https://sqoop.apache.org/), to create complex queries and map attributes to index based on the result.

Learn more at https://docs.viglet.com/turing/connectors/#database

4.1.3. File System

Command line to index files, extracting text from files such as Word, Excel, PDF, including images, through OCR.

Learn more at https://docs.viglet.com/turing/connectors/#file-system

4.1.4. OpenText WEM Listener

OpenText WEM Listener to publish content to Viglet Turing.

Learn more at https://docs.viglet.com/turing/connectors/#wem

4.1.5. Wordpress

Wordpress plugin that allows you to index posts.

Learn more at https://docs.viglet.com/turing/connectors/#wordpress

4.2. Named Entity Recognition (NER)

With NLP it is possible to detect entities such as:

- People
- Places
- Organizations
- Money
- Time
- Percentage

4.3. Facets

Define attributes that will be used as filters for your navigation, consolidating the total content in your display

4.4. Targeting Rules

Through attributes defined in the contents, it is possible to use them to restrict their display based on the user's profile.

4.5. SDK Java

Java API (https://github.com/openturing/turing-java-sdk) facilitates the use and access to Viglet Turing ES, without the need for consumer search content with complex queries.

Chapter 5. Chatbot

Communicate with your client and elaborate complex intents, obtain reports and progressively evolve your interaction.

Its components:

5.1. Agent

Handles conversations with your end users. It is a natural language processing module that understands the nuances of human language

5.2. Intent

An intent categorizes an end user’s intention for taking a conversation shift. For each agent, you define several intents, where your combined intents can handle a complete conversation.

5.3. Actions

The field of action is a simple field of convenience that helps to execute logic in the service.

5.4. Entity

Each intent parameter has a type, called an entity type, that dictates exactly how the data in an end user expression is extracted.

5.5. Training

Defines and corrects intents.

5.6. History

Shows the conversation history and reports.

Chapter 6. OpenText Blazon Integration

Turing ES detects Entities of OpenText Blazon Documents using OCR and NLP, generating Blazon XML to show the entities into document.

Chapter 7. Turing ES Console

Turing ES has many components: Search Engine, NLP, Converse (Chatbot), Semantic Navigation

7.1. Login

When access the Turing ES, appear a login page. For default the login/password is `admin` /`admin`

[Login] | *img/screenshots/turing-login.png*

Figure 2. Login Page

7.2. Search Engine

7.2.1. Configuration

Search Engine is used by Turing to store and retrieve data of Converse (Chatbot) and Semantic Navigation Sites.

[Search Page] | *img/screenshots/turing-se.png*

Figure 3. Search Engine Page

It is possible create or edit a Search Engine with following attributes:

Table 1. Search Engine Attributes

Attribute	Description
Name	Name of Search Engine
Description	Description of Search Engine
Vendor	Select the Vendor of Search Engine. For now it only supports Solr.
Host	Host name where the Search Engine service is installed
Port	Port of Search Engine Service
Enabled	If the Search Engine is enabled.

7.3. Semantic Navigation

7.3.1. Configuration

[Semantic Navigation Page] | *img/screenshots/turing-sn.png*

Figure 4. Semantic Navigation Page

Settings Tab

The Settings of Semantic Navigation Site contains the following attributes:

Table 2. Semantic Navitation Site Settings

Attribute	Description
Name	Name of Semantic Navigation Site.
Description	Description of Semantic Navigation Site.
Search Engine	Select the Search Engine that was created in Search Engine Section. The Semantic Navigation Site will use this Search Engine to store and retrieve data.
NLP Vendor	NLP Vendor for this site.
Thesaurus	If will use Thesaurus.

Multi Languages Tab

The Multi Languages of Semantic Navigation Site contains the following attributes:

Table 3. Multi Language Settings

Attribute	Description
Language	Language for Semantic Navigation SIte.
NLP Instance	NLP Instance to detect entities during indexing.

Attribute	Description
Core	Solr Core Name to store and to search indexed content.

Behavior Tab

Contains the following attributes:

Table 4. Semantic Navitation Site Appearance Attributes

Section	Attribute	Description
Behavior	Number of items per page	Number of items that will appear in search.
Facet	Facet enabled	If it will be show Facet (Filters) on search.
	Number of items per facet	Number of items that will appear in each Facet (Filter).
Highlighting	Highlighting enabled	Define whether to show highlighted lines.
	Pre Tag	HTML Tag that will be used on begin of term. For example: <mark>
	Post Tag	HTML Tag that will be used on the end of term. For example: </mark>
Did you mean?	"Did you mean?" enabled	Use "did you mean?" feature.
	Always show the search with the corrected term.	If the term is misspelled, it already shows the search with the corrected term. If disabled, it shows the search with the entered term in the search.
MLT	More Like This enabled?	Define whether to show MLT

Section	Attribute	Description
Default Fields	Title	Field that will be used as title that is defined in Solr schema.xml
	Text	Field that will be used as title that is defined in Solr schema.xml
	Description	Field that will be used as description that is defined in Solr schema.xml
	Date	Field that will be used as date that is defined in Solr schema.xml
	Image	Field that will be used as Image URL that is defined in Solr schema.xml
	URL	Field that will be used as URL that is defined in Solr schema.xml

Merge Providers Details Tab

Merge Providers Details Tab contains the following attributes: .Semantic Navitation Site Merge Providers Attributes

Section	Attribute	Description
Providers	Source	Name of Source Provider.
	Destination	Name of Destination Provider.
Relations	Source	Relation Identifier of Source Provider.
	Destination	Relation Identifier of Destination Provider.
Description	Description	More about merge providers.
Overwritten Fields	Name	Name of Source Field that overwritten destination field or create new one.

Fields Tab

Fields Tab contains a table with the following columns: .Semantic Navitation Site Fields Columns

Column Name	Description
Type	Type of Field. It can be: - NER (Named Entity Recognition) used by NLP. - Seach Engine used by Solr.
Field	Name of Field.
Enabled	If the field is enabled or not.
MLT	If this field will be used in MLT.
Facets	To use this field like a facet (filter)
Highlighting	If this field will show highlighted lines.
NLP	If this field will be processed by NLP to detect Entities (NER) like People, Organization and Place.

When click in Field appear a new page with Field Details with the following attributes:

Table 5. Semantic Navitation Site Fields Detail Attributes

Attribute	Description
Name	Name of Field
Description	Description of Field
Type	Type of Field. It can be: `INT`, `LONG`, `STRING`, `DATE` and `BOOL`
Multi Valued	If is a array

Attribute	Description
Facet Name	Name of Label of Facet (Filter) on Search Page.
Facet	To use this field like a facet (filter)
Highlighting	If this field will show highlighted lines.
MLT	If this field will be used in MLT.
Enabled	If the field is enabled.
Required	If the field is required.
Default Value	Case the content is indexed without these field, that is the default value.
NLP	If this field will be processed by NLP to detect Entities (NER) like People, Organization and Place.

Spotlight Details Tab

Spotlight Details Tab contains the following attributes:

Table 6. Semantic Navigation Site Spotlight Attributes

Attribute	Description
Name	Spotlight Name
Description	Spotlight Description
Terms	If any of these terms are searched for, this will trigger documents to display as spotlights.
Indexed Documents	These documents will display as spotlights when there are search terms.

7.3.2. Search Page

HTML

In `Turing ES Console` > `Semantic Navigation` > `<SITE_NAME>` > `Multi languages` > click in `Open Search` button of some language.

It will open a Search Page that uses the pattern:

```
GET http://localhost:2700/sn/<SITE_NAME>
```

JSON

This page requests the Turing Rest API via AJAX. For example, to return all results of Semantic Navigation Site in JSON Format:

```
GET http://localhost:2700/api/sn/<SITE_NAME>/search?p=1&q=*&sort=relevance
```

Table 7. Semantic Navigation Rest API Get Attributes

Attribute	Required / Optional	Description	Example
q	Required	Search Query.	q=foo
p	Required	Page Number, first page is 1.	p=1
sort	Required	Sort values: `relevance`, `newest` and `oldest`.	sort=relevance
fq[]	Optional	Query Field. Filter by field, using the following pattern: **FIELD**: **VALUE**.	fq[]=title:bar

Attribute	Required / Optional	Description	Example
tr[]	Optional	Targeting Rule. Restrict search based in: **FIELD**: **VALUE**.	tr[]=department:foobar
rows	Optional	Number of rows that query will return.	rows=10

Chapter 8. Customer Case Studies

8.1. Brazil

8.1.1. Insurance Company

On Intranet of Insurance Company uses OpenText WEM and OpenText Portal integrated with Dynamic Portal Module, a consolidated search was created in Viglet Turing ES, using the connectors: WEM, Database with File System. In this way it was possible to display all the contents and files of the search Intranet, with targeting rules, allowing only to display content that the user has permission. The OpenText Portal accesses Viglet Turing ES Java API, so it was not necessary to create complex queries to return the results.

8.1.2. Government Company

A set of API Rest was created to make all Government Company content available to partners. All these contents are in OpenText WEM and the WEM connector was used to index the contents on Viglet Turing ES. A Spring Boot application was created with the Rest API set that consumes Turing ES content through the Viglet Turing ES Java API.

8.1.3. University

Brazilian University website was developed using Viglet Shio CMS (https://viglet.com/shio), and all contents are indexed in Viglet Turing ES automatically. This configuration was made in content modeling and the development of the search template was made in Viglet Shio CMS.

8.2. Spanish

8.2.1. Insurance Company

Institutional website of the insurance company using OpenText Dynamic Site Module with multi languages, an integrated search was created with Turing ES, using connectors: WEM and Apache Nutch.